DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture LAB
EXPERIMENT # 08-09

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Score: __________ Signature of the Lab Tutor: ____________ Date:

Introduction to HDLC, PPP,PAP and CHAP

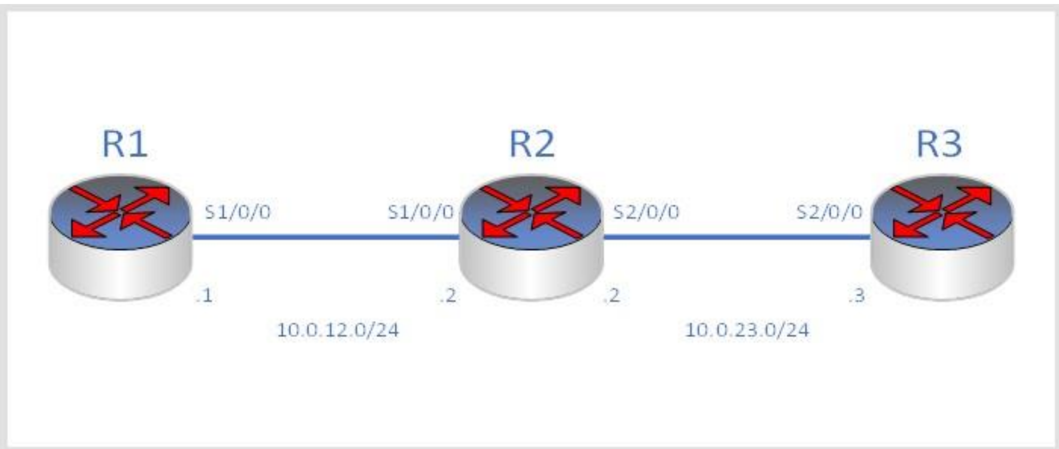
LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Establish HDLC encapsulation as the serial link layer protocol.
- Change the DCE clock baud rate on a serial link.
- Establish PPP encapsulation as the serial link layer protocol.
- Implementation of PAP authentication on the PPP link.
- Implementation of CHAP authentication on the PPP link.

Topology



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Scenario

As an expanding enterprise business, multiple branch offices have been established and are to be part of the company's administrative domain. WAN solutions are required and as the network administrator the company you have been tasked with establishing HDLC and PPP solutions at the edge router to be carried over some service provider network, possibly MPLS, however the details of this have not been revealed to you since the service provider network remains outside of the scope of your task. R2 is an edge router located in the HQ, and R1 and R3 are located in branch offices. The HQ and branches need to be established as a single administrative domain. Use HDLC and PPP on the WAN links, and establish authentication as a simple security measure.

Tasks

Step 1 Preparing the environment

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R1

<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R2

<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R3
```

Step 2 Configure serial interface IP addressing for R1, R2 & R3

```
[R1] interface Serial 1/0/0
[R1-Serial1/0/0] ip address 10.0.12.1 24

[R2] interface Serial 1/0/0
[R2-Serial1/0/0] ip address 10.0.12.2 24
[R2-Serial1/0/0] quit
[R2] interface Serial 2/0/0
[R2-Serial2/0/0] ip address 10.0.23.2 24

[R3] interface Serial 2/0/0
[R3-Serial2/0/0] ip address 10.0.23.3 24
```

Step 3 Enable the HDLC protocol on the serial interfaces

```
[R1] interface Serial 1/0/0
[R1-Serial1/0/0] link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R2] interface Serial 1/0/0
[R2-Serial1/0/0] link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R2-Serial1/0/0] quit
[R2] interface Serial 2/0/0
[R2-Serial2/0/0] link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R3] interface Serial 2/0/0
[R3-Serial2/0/0] link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
```

After HDLC is enabled on the serial interfaces, view the serial interface status. The displayed information for R1 should be used as an example.

```
[R1] display interface Serial1/0/0
Serial1/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2016-3-10 11:25:08
Description: HUAWEI, AR Series, Serial1/0/0 Interface
Route Port, The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.12.1/24
Link layer protocol is nonstandard HDLC
Last physical up time   : 2016-3-10 11:23:55
Last physical down time : 2016-3-10 11:23:55
Current system time: 2016-3-10 11:25:46
Physical layer is synchronous, Baudrate is 64000 bps
Interface is DCE, Cable type is V24, Clock mode is DCECLK
Last 300 seconds input rate 3 bytes/sec 24 bits/sec 0 packets/sec  Last
300 seconds output rate 3 bytes/sec 24 bits/sec 0 packets/sec

Input: 100418 packets, 1606804 bytes
  Broadcast:             0,  Multicast:                      0
Errors:                 0,  Runts:                          0
Giants:                0,  CRC:                           0
```

```
  Alignments:              0,  Overruns:                        0
Dribbles:                0,  Aborts:                        0    No
Buffers:             0,  Frame Error:                    0

Output: 100418 packets, 1606830 bytes
  Total Error:             0,  Overruns:                        0
Collisions:              0,  Deferred:                      0
  No Buffers:              0
DCD=UP DTR=UP DSR=UP RTS=UP CTS=UP
    Input bandwidth utilization  : 0.06%
    Output bandwidth utilization : 0.06%
```

Test connectivity of the directly connected link after verifying that the physical status and protocol status of the interface are Up.

```
<R2> ping 10.0.12.1
  PING 10.0.12.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=44 ms
Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=39 ms      Reply
from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=39 ms      Reply from
10.0.12.1: bytes=56 Sequence=4 ttl=255 time=40 ms      Reply from
10.0.12.1: bytes=56 Sequence=5 ttl=255 time=39 ms

  --- 10.0.12.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 39/40/44 ms

[R2] ping 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=44 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=39 ms      Reply
from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=39 ms      Reply from
10.0.23.3: bytes=56 Sequence=4 ttl=255 time=40 ms      Reply from
10.0.23.3: bytes=56 Sequence=5 ttl=255 time=39 ms

  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
    round-trip min/avg/max = 39/40/44 ms
```

Step 4 Configure Routes (Static)

You can skip this step, if you want to configure routes dynamically.

```
<R1> display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 7        Routes : 7

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      10.0.12.0/24  Direct  0    0           D   10.0.12.1       Serial1/0/0
      10.0.12.1/32  Direct  0    0           D   127.0.0.1       Serial1/0/0
    10.0.12.255/32  Direct  0    0           D   127.0.0.1       Serial1/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

<R1> system-view
[R1]ip route-static 0.0.0.0 0 s1/0/0  [R3]ip
route-static 0.0.0.0 0 s2/0/0
```

After the configuration is complete, check that all the routes have been learned.

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0           D   10.0.12.1       Serial1/0/0
      10.0.12.0/24  Direct  0    0           D   10.0.12.1       Serial1/0/0
      10.0.12.1/32  Direct  0    0           D   127.0.0.1       Serial1/0/0
    10.0.12.255/32  Direct  0    0           D   127.0.0.1       Serial1/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

```
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
[R3]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0           D   10.0.23.2       Serial2/0/0
      10.0.23.0/24  Direct  0    0           D   10.0.23.2       Serial2/0/0
      10.0.23.2/32  Direct  0    0           D   127.0.0.1       Serial2/0/0
    10.0.23.255/32  Direct  0    0           D   127.0.0.1       Serial2/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

On R1, run the **ping** command to test connectivity between R1 and R3.

```
<R1>ping 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=44 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=39 ms      Reply
from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=39 ms      Reply from
10.0.23.3: bytes=56 Sequence=4 ttl=254 time=40 ms      Reply from
10.0.23.3: bytes=56 Sequence=5 ttl=254 time=39 ms

  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
    round-trip min/avg/max = 39/40/44 ms
```

Step 5 Configure RIPv2 (Dynamic)

Skip this step, if you have configured routes in step 4.

Enable the RIP routing protocol to advertise the remote networks of R1 & R3.

```
[R1]rip
[R1-rip-1]version 2
```

```
[R1-rip-1]network 10.0.0.0

[R2]rip
[R2-rip-1]version 2
[R2-rip-1]network 10.0.0.0

[R3]rip
[R3-rip-1]version 2
[R3-rip-1]network 10.0.0.0
```

After the configuration is complete, check that all the routes have been learned. Verify that corresponding routes are learned by RIP.

```
<R1>display ip routing-table

Route Flags: R - relay, D - download to fib
----------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost   Flags NextHop      Interface

10.0.12.0/24        Direct  0    0       D    10.0.12.1    Serial1/0/0
10.0.12.1/32        Direct  0    0       D    127.0.0.1    Serial1/0/0
10.0.12.255/32      Direct  0    0       D    127.0.0.1    Serial1/0/0
10.0.23.0/24        RIP     100  1       D    10.0.12.2    Serial1/0/0
127.0.0.0/8         Direct  0    0       D    127.0.0.1    InLoopBack0  127.0.0.1/32
Direct  0    0      D    127.0.0.1    InLoopBack0  127.255.255.255/32  Direct  0
0      D    127.0.0.1    InLoopBack0  255.255.255.255/32  Direct  0    0      D
127.0.0.1    InLoopBack0
```

On R1, run the ping command to test connectivity between R1 and R3.

```
<R1>ping 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=44 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=39 ms      Reply
from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=39 ms      Reply from
10.0.23.3: bytes=56 Sequence=4 ttl=254 time=40 ms      Reply from
10.0.23.3: bytes=56 Sequence=5 ttl=254 time=39 ms
```

```
  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
    round-trip min/avg/max = 39/40/44 ms
```

Step 6 Manage the serial connection

View the type of the cable connected to the serial interface, interface status, and clock frequency, and change the clock frequency.

```
<R1>display interface Serial1/0/0
Serial1/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2016-03-10 11:25:08
Description:HUAWEI, AR Series, Serial1/0/0 Interface
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.12.1/24
Link layer protocol is nonstandard HDLC
Last physical up time   : 2016-03-10 11:23:55
Last physical down time : 2016-03-10 11:23:55
Current system time: 2016-03-10 11:51:12
Physical layer is synchronous, Baudrate is 64000 bps
Interface is DCE, Cable type is V35, Clock mode is DCECLK1
Last 300 seconds input rate 5 bytes/sec 40 bits/sec 0 packets/sec  Last
300 seconds output rate 2 bytes/sec 16 bits/sec 0 packets/sec
…output omit…
```

The preceding information shows that S1/0/0 on R1 connects to a DCE cable and the clock frequency is 64000 bit/s. The DCE controls the clock frequency and bandwidth.

Change the clock frequency on the link between R1 and R2 to 128000 bit/s. This operation must be performed on the DCE, R1.

```
[R1]interface Serial 1/0/0
[R1-Serial1/0/0]baudrate 128000
```

After the configuration is complete, view the serial interface status.

```
<R1>display interface Serial1/0/0
Serial1/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2016-03-10 11:25:08
Description:HUAWEI, AR Series, Serial1/0/0 Interface
```

```
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.12.1/24
Link layer protocol is nonstandard HDLC
Last physical up time   : 2016-03-10 11:23:55
Last physical down time : 2016-03-10 11:23:55
Current system time: 2016-03-10 11:54:19
Physical layer is synchronous, Baudrate is 128000 bps
Interface is DCE, Cable type is V35, Clock mode is DCECLK1
Last 300 seconds input rate 6 bytes/sec 48 bits/sec 0 packets/sec  Last
300 seconds output rate 4 bytes/sec 32 bits/sec 0 packets/sec
…output omit…
```

Step 7 Configure PPP on the serial interfaces

Configure PPP between R1 and R2, as well as R2 and R3. Both ends of the link must use the same encapsulation mode. If different encapsulation modes are used, interfaces may display as 'Down'.

```
[R1]interface Serial 1/0/0
[R1-Serial1/0/0]link-protocol ppp
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R2]interface Serial 1/0/0
[R2-Serial1/0/0]link-protocol ppp
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R2-Serial1/0/0]quit
[R2]interface Serial 2/0/0
[R2-Serial2/0/0]link-protocol ppp
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R3]interface Serial 2/0/0
[R3-Serial2/0/0]link-protocol ppp
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
```

After the configuration is complete, test link connectivity.

```
<R2>ping 10.0.12.1
  PING 10.0.12.1: 56  data bytes, press CTRL_C to break
    Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=22 ms
Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=27 ms      Reply
from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=27 ms      Reply from
10.0.12.1: bytes=56 Sequence=4 ttl=255 time=27 ms      Reply from
10.0.12.1: bytes=56 Sequence=5 ttl=255 time=27 ms

  --- 10.0.12.1 ping statistics ---
```

```
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
    round-trip min/avg/max = 22/26/27 ms

<R2>ping 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=35 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=40 ms      Reply
from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=40 ms      Reply from
10.0.23.3: bytes=56 Sequence=4 ttl=255 time=40 ms      Reply from
10.0.23.3: bytes=56 Sequence=5 ttl=255 time=40 ms

  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 35/39/40 ms
```

If the **ping** operation fails, check the interface status and whether the link layer protocol type is correct.

```
<R1>display interface Serial1/0/0
Serial1/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2016-03-10 12:35:41
Description: HUAWEI, AR Series, Serial1/0/0 Interface
Route Port, The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.12.1/24
Link layer protocol is PPP
LCP opened, IPCP opened
Last physical up time   : 2016-03-10 11:57:20
Last physical down time : 2016-03-10 11:57:19
Current system time: 2016-03-10 13:38:03
Physical layer is synchronous, Baudrate is 128000 bps
Interface is DCE, Cable type is V35, Clock mode is DCECLK1
Last 300 seconds input rate 7 bytes/sec 56 bits/sec 0 packets/sec  Last
300 seconds output rate 4 bytes/sec 32 bits/sec 0 packets/sec
…output omit…
```

Step 8 Check routing entry changes

After PPP configuration is complete, routers establish connections at the data link layer. The local device sends a route to the peer device. The route contains the interface IP address and a 32-bit mask. The following information uses R2 as an example, for which the routes to R1 and R3 can be seen.

```
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
----------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 12       Routes : 12

Destination/Mask    Proto   Pre  Cost  Flags  NextHop             Interface
10.0.12.0/24        Direct  0    0     D      10.0.12.2           Serial1/0/0
10.0.12.1/32        Direct  0    0     D      10.0.12.1           Serial1/0/0
10.0.12.2/32        Direct  0    0     D      127.0.0.1           Serial1/0/0
10.0.12.255/32      Direct  0    0     D      127.0.0.1           Serial1/0/0
10.0.23.0/24        Direct  0    0     D      10.0.23.2           Serial2/0/0
10.0.23.2/32        Direct  0    0     D      127.0.0.1           Serial2/0/0
10.0.23.3/32        Direct  0    0     D      10.0.23.3           Serial2/0/0
10.0.23.255/32      Direct  0    0     D      127.0.0.1           Serial2/0/0
```

```
127.0.0.0/8          Direct  0    0     D      127.0.0.1            InLoopBack0
127.0.0.1/32         Direct  0    0     D      127.0.0.1            InLoopBack0
127.255.255.255/32   Direct  0    0     D      127.0.0.1            InLoopBack0
255.255.255.255/32   Direct  0    0     D      127.0.0.1            InLoopBack0
```

Think about the origin and functions of the two routes. Check the following items:

If HDLC encapsulation is used, do these two routes exist?

Can R1 and R2 communicate using HDLC or PPP when the IP addresses of S1/0/0 interfaces on R1 and R2 are located on different network segments?

Step 9 Enable PAP authentication between R1 and R2

Configure PAP authentication with R1 as the PPP PAP authenticator.

```
[R1]interface Serial 1/0/0
[R1-Serial1/0/0]ppp authentication-mode pap
[R1-Serial1/0/0]quit
[R1]aaa
[R1-aaa]local-user huawei password cipher huawei123
info: A new user added
[R1-aaa]local-user huawei service-type ppp
```

Configure PAP authentication with R2 acting as the PAP authenticated device.

```
[R2]interface Serial 1/0/0
[R2-Serial1/0/0]ppp pap local-user huawei password cipher huawei123
```

After R2 sends an authentication request to R1, R1 sends a response message to R2, requesting R2 to use PAP authentication following which R2 will send its password to R1.

After the configuration is complete, test connectivity between R1 and R2.

```
<R1>debugging ppp pap packet
<R1>terminal debugging  <R1>display
debugging
PPP PAP packets debugging switch is on
<R1>system-view
[R1]interface Serial 1/0/0
[R1-Serial1/0/0]shutdown
[R1-Serial1/0/0]undo shutdown

Mar 10 2016 14:44:22.440.1+00:00 R1 PPP/7/debug2:
```

```
  PPP Packet:
      Serial1/0/0 Input  PAP(c023)Pkt, Len 22
      State ServerListen, code Request(01), id 1, len 18
Host Len:  6 Name:huawei
[R1-Serial1/0/0]
Mar 10 2016 14:44:22.440.2+00:00 R1 PPP/7/debug2:
  PPP Packet:
      Serial1/0/0 Output PAP(c023)Pkt, Len 52
      State WaitAAA, code Ack(02), id 1, len 48
      Msg Len: 43  Msg:Welcome to use Access ROUTER, Huawei Tech.

[R1-Serial1/0/0]return
<R1>undo debugging all
Info: All possible debugging has been turned off
```

Step 10 Enable CHAP authentication between R2 and R3

Configure R3 as the authenticator. After R2 sends an authentication request to R3, R3 sends a response message to R2, requesting R2 to use CHAP authentication following which a challenge is sent to R3.

```
[R3]interface Serial 2/0/0
[R3-Serial2/0/0]ppp authentication-mode chap
[R3-Serial2/0/0]quit
[R3]aaa
[R3-aaa]local-user huawei password cipher huawei123
info: A new user added  [R3-aaa]local-user huawei
service-type ppp
[R3-aaa]quit
[R3]interface Serial 2/0/0
[R3-Serial2/0/0]shutdown
[R3-Serial2/0/0]undo shutdown
```

On R3, the following information is displayed.

```
Dec 10 2013 15:06:00+00:00 R3 %%01PPP/4/PEERNOCHAP(l)[5]:On the interface Serial2/0/0, authentication failed and PPP link was closed because CHAP was disabled on the peer.
[R3-Serial2/0/0]
Dec 10 2013 15:06:00+00:00 R3 %%01PPP/4/RESULTERR(l)[6]:On the interface Serial2/0/0, LCP negotiation failedbecause the result cannot be accepted.
```

The highlighted output indicates that authentication is unable to initialize. Configure R2 as the CHAP client.

```
[R2]interface Serial 2/0/0
[R2-Serial2/0/0]ppp chap user huawei
[R2-Serial2/0/0]ppp chap password cipher huawei123
```

After the configuration is complete, the interface changes to an Up state. The ping command output is as follows:

```
<R2>ping 10.0.23.3
  PING 10.0.23.3: 56  data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=35 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=41 ms      Reply
from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=41 ms      Reply from
10.0.23.3: bytes=56 Sequence=4 ttl=255 time=41 ms      Reply from
10.0.23.3: bytes=56 Sequence=5 ttl=255 time=41 ms

  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
0.00% packet loss
    round-trip min/avg/max = 35/39/41 ms
```

Step 11 PPP CHAP debugging

Run the debug command to view negotiation of the PPP connection between R2 and R3. The PPP connection is established using CHAP. Disable interface Serial 2/0/0 on R2, run the debug command, and enable Serial 2/0/0 on R2.

```
[R2]interface Serial 2/0/0
[R2-Serial2/0/0]shutdown
```

Run the **debugging ppp chap all** and the **terminal debugging** commands to display the debugging information.

```
[R2-Serial2/0/0]return
<R2>debugging ppp chap all
<R2>terminal debugging
Info: Current terminal debugging is on.
<R2>display debugging
```

```
PPP CHAP packets debugging switch is on
PPP CHAP events debugging switch is on  PPP
CHAP errors debugging switch is on
PPP CHAP state change debugging switch is on
```

Force CHAP authentication to initialize on S2/0/0 of R2.

```
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]interface Serial 2/0/0
[R2-Serial2/0/0]undo shutdown
```

The following debugging information is displayed:

```
Mar 10 2016 09:10:38.700.1+00:00 R2 PPP/7/debug2:
  PPP State Change:
      Serial2/0/0 CHAP : Initial --> ListenChallenge
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.710.1+00:00 R2 PPP/7/debug2:
  PPP Packet:
      Serial2/0/0 Input  CHAP(c223)Pkt, Len 25
      State ListenChallenge, code Challenge(01), id 1, len 21
      Value_Size:  16 Value: fc 9b 56 e1 53 e3 a6 26 1b 54 e5 e2 a1 ed 90 87
Name:
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.710.2+00:00 R2 PPP/7/debug2:
  PPP Event:
      Serial2/0/0 CHAP Receive Challenge Event
state ListenChallenge
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.710.3+00:00 R2 PPP/7/debug2:
  PPP Packet:
      Serial2/0/0 Output CHAP(c223)Pkt, Len 31
      State ListenChallenge, code Response(02), id 1, len 27
      Value_Size:  16 Value: f9 54  1 69 30 59 a0 af 52 a1 1d de 85 77 27 6b
      Name: huawei
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.710.4+00:00 R2 PPP/7/debug2:
  PPP State Change:
      Serial2/0/0 CHAP : ListenChallenge --> SendResponse
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.720.1+00:00 R2 PPP/7/debug2:
```

```
  PPP Packet:
      Serial2/0/0 Input  CHAP(c223)Pkt, Len 20
State SendResponse, code SUCCESS(03), id 1, len 16
Message: Welcome to .
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.720.2+00:00 R2 PPP/7/debug2:
  PPP Event:
      Serial2/0/0 CHAP Receive Success Event
state SendResponse
[R2-Serial2/0/0]
Mar 10 2016 09:10:38.720.3+00:00 R2 PPP/7/debug2:
  PPP State Change:
      Serial2/0/0 CHAP : SendResponse --> ClientSuccess
```

The highlighted debugging information shows the key CHAP behavior. Disable the debugging process.

```
[R2-Serial2/0/0]return
<R2>undo debugging all
Info: All possible debugging has been turned off
```

Final Configuration

```
[R1]display current-configuration
[V200R007C00SPC600]
#   sysname
R1
#  aaa
 authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default    domain
default_admin
 local-user admin password cipher %$%$=i~>Xp&aY+*2cEVcS-A23Uwe%$%$   local-
user admin service-type http
 local-user huawei password cipher %$%$B:%I)Io0H8)[%SB[idM3C/!#%$%$   local-
user huawei service-type ppp
#
interface Serial1/0/0   link-
protocol ppp   ppp authentication-
mode pap    ip address 10.0.12.1
255.255.255.0    baudrate 128000
```

```
#  rip 1
version 2
network 10.0.0.0
#
user-interface con 0   authentication-
mode password
 set authentication password cipher %$%$dD#}P<HzJ;Xs%X>hOkm!,.+Iq61QK`K6tI}cc-
;k_o`C.+L,%$%$  user-interface
vty 0 4
#
return

[R2]display current-configuration
[V200R007C00SPC600]
#   sysname
R2  #
interface Serial1/0/0   link-
protocol ppp
 ppp pap local-user huawei password cipher %$%$u[hr6d<JVHR@->T7xr1<$.iv%$%$
ip address 10.0.12.2 255.255.255.0
#
interface Serial2/0/0
link-protocol ppp   ppp
chap user huawei
 ppp chap password cipher %$%$e{5h)gh"/Uz0mUC%vEx3$4<m%$%$
ip address 10.0.23.2 255.255.255.0
#  rip 1
version 2
network 10.0.0.0
#
user-interface con 0
authentication-mode password   set
authentication password cipher
%$%$|nRPL^hr2IXi7LHDID!/,.*%.8%h;3:,hXO2dk#ikaWI.*(,%$%$  user-interface
vty 0 4
#
return

[R3]display current-configuration
[V200R007C00SPC600]
```

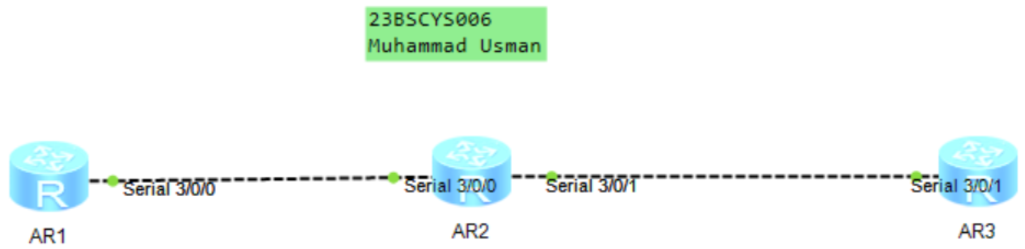
```
#   sysname
R3  #  aaa
 authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default    domain
default_admin
 local-user admin password cipher %$%$=i~>Xp&aY+*2cEVcS-A23Uwe%$%$   local-
user admin service-type http
 local-user huawei password cipher %$%$fZsyUk1=O=>:L4'ytgR~D*Im%$%$   local-
user huawei service-type ppp
#
interface Serial2/0/0   link-
protocol ppp   ppp authentication-
mode chap    ip address 10.0.23.3
255.255.255.0
#  rip 1
version 2
network 10.0.0.0
#
user-interface con 0
authentication-mode password   set
authentication password cipher
%$%$W|$)M5D}v@bY^gK\;>QR,.*d;8Mp>|+EU,:~D~8b59~..*g,%$%$  user-interface
vty 0 4
#  return
```

Tasks

Try a different topology and demonstrate your learning.

Upload the `.topo` `.efz` `.cfg` files

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture LAB
EXPERIMENT # 08-09



```
[23BSCYS006-R1]dis current-configuration
[V200R003C00]
#
 sysname 23BSCYS006-R1
#
 board add 0/3 2SA
 board add 0/4 2SA
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
 local-user huawei password cipher %$%$DF@MC7b,zB<J9r=bJaqUYGQD%$%$
 local-user huawei service-type ppp
#
firewall zone Local
 priority 15
#
```

```
#
interface Serial3/0/0
 link-protocol ppp
 ppp authentication-mode pap
 ip address 10.0.12.1 255.255.255.0
 virtualbaudrate 128000
#
interface Serial3/0/1
 link-protocol ppp
#
interface Serial4/0/0
 link-protocol ppp
#
interface Serial4/0/1
 link-protocol ppp
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
rip 1
 version 2
 network 10.0.0.0
#
ip route-static 0.0.0.0 0.0.0.0 Serial3/0/0
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[23BSCYS006-R1]
```

```
#
 sysname 23BSCYS006-R2
#
 board add 0/3 2SA
 board add 0/4 2SA
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
firewall zone Local
 priority 15
#
interface Serial3/0/0
 link-protocol ppp
 ppp pap local-user huawei password cipher %$%$o{n$;R"UDXf}-*Y*#zP6,#P;%$%$
 ip address 10.0.12.2 255.255.255.0
#
```

```
#
interface Serial3/0/0
 link-protocol ppp
 ppp pap local-user huawei password cipher %$%$o{n$;R"UDXf}-*Y*#zP6,#P;%$%$
 ip address 10.0.12.2 255.255.255.0
#
interface Serial3/0/1
 link-protocol ppp
 ppp chap user huawei
 ppp chap password cipher %$%$)[XJ96-:`9@f#M3v]HaR,##q%$%$
 ip address 10.0.23.2 255.255.255.0
#
interface Serial4/0/0
 link-protocol ppp
#
interface Serial4/0/1
 link-protocol ppp
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
rip 1
 version 2
 network 10.0.0.0
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[23BSCYS006-R2]
```

```
[23BSCYS006-R3]dis current-configuration
[V200R003C00]
#
 sysname 23BSCYS006-R3
#
 board add 0/3 2SA
 board add 0/4 2SA
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
 drop illegal-mac alarm
#
 wlan ac-global carrier id other ac id 0
#
 set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
 local-user huawei password cipher %$%$[sz_)=g0RNzQ3wY%7h@JYI*b%$%$
 local-user huawei service-type ppp
#
firewall zone Local
 priority 15
#
```

```
interface Serial3/0/0
 link-protocol ppp
#
interface Serial3/0/1
 link-protocol ppp
 ppp authentication-mode chap
 ip address 10.0.23.3 255.255.255.0
#
interface Serial4/0/0
 link-protocol ppp
#
interface Serial4/0/1
 link-protocol ppp
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
rip 1
 version 2
 network 10.0.0.0
#
ip route-static 0.0.0.0 0.0.0.0 Serial3/0/1
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[23BSCYS006-R3]
```